INCH-POUND

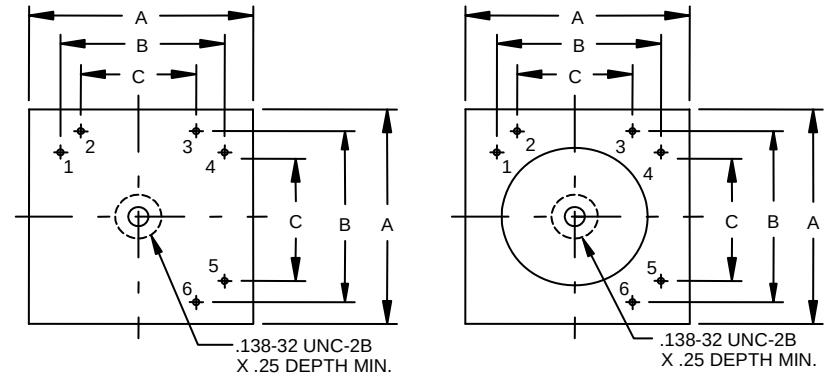
MIL-PRF-27/43C
18 January 2010
SUPERSEDING
MIL-PRF-27/43B
7 February 1984

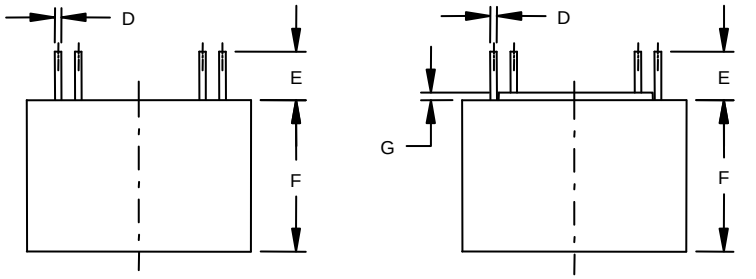
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 9 VOLT AMPERES, 400 HZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.





CONFIGURATION A

CONFIGURATION B

"X" VOLTAGE CT

"X" mA OR A

MAX W.V. , "X"V"
MAX ALT 70 .000

(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING

Dim	Inches	Millimeters
A	1.09 Max	27.2
B	.800	20.32
C	.600	15.24
D	.032 ±.002	0.81 ±0.05
E	.25 .13	6.4 3.3
F	1.03 Max	26.2
G	.03 .02	0.8 0.5

FIGURE 1. Dimensions and configurations

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm).
4. Marking shall be on the side of the case.
5. A white dot shall be marked on the side opposite the terminals, over terminal 1, in addition to the terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configurations – continued.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 2.2 ounces, maximum.

Duty cycle: Continuous.

Case: Plastic encapsulated.

Material: Plastic.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55˚C to +130˚C, maximum.

Terminals: Printed circuit pins – tinned brass.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts rms, 400 Hz across (1-2): Current in (1-2) shall not exceed 9 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a restrictive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 83 percent minimum at rated load.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{\text{Voltage (no Load) - Voltage (rated load)}}{\text{Voltage (rated load)}} \times 100$

shall not exceed 10 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 30˚C, maximum.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.

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VERIFICATION

Extent of qualification

Qualification testing and approval to M27/37-44 A or B, M27/42-44 A or B, and M27/46-44 A or B shall be sufficient to grant qualification approval to MIL-PRF-27/37 through MIL-PRF-27/47 inclusive, all parts.

Qualification testing and approval to M27/43-44 shall be sufficient to grant qualification approval to M27/43-01 through M27/43-44.

Part or Identification Number (PIN): M27/43-01 A or B.

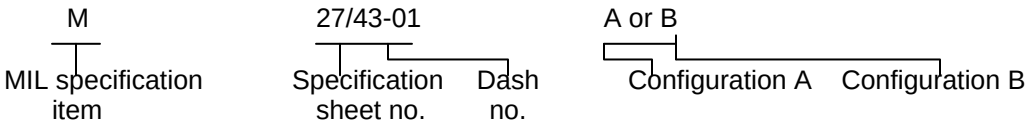

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TABLE I. Electrical ratings.

Dash No.	Type Designation	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms) (MAX)
01 A or B	664	6	1.500	325
02 A or B	665	6.3	1.430	325
03 A or B	666	7	1.286	325
04 A or B	667	7.5	1.200	325
05 A or B	668	8	1.125	325
06 A or B	669	9	1.000	325
07 A or B	670	10	0.900	325
08 A or B	671	11	0.818	325
09 A or B	672	12	0.750	325
10 A or B	673	12.6	0.715	325
11 A or B	674	13	0.692	325
12 A or B	675	15	0.600	325
13 A or B	676	16	0.563	325
14 A or B	677	18	0.500	325
15 A or B	678	20	0.450	325
16 A or B	679	22	0.409	325
17 A or B	680	24	0.375	325
18 A or B	681	26	0.346	325
19 A or B	682	28	0.321	325
20 A or B	683	30	0.300	325
21 A or B	684	33	0.273	325
22 A or B	685	36	0.250	325
23 A or B	686	39	0.231	325
24 A or B	687	43	0.209	325
25 A or B	688	47	0.191	325
26 A or B	689	51	0.176	325
27 A or B	690	56	0.161	325
28 A or B	691	62	0.145	325
29 A or B	692	68	0.132	325
30 A or B	693	75	0.120	325
31 A or B	694	82	0.110	325
32 A or B	695	91	0.099	325
33 A or B	696	100	0.090	325
34 A or B	697	110	0.082	325
35 A or B	698	115	0.078	325
36 A or B	699	120	0.075	325
37 A or B	700	130	0.069	325
38 A or B	701	150	0.060	325
39 A or B	702	180	0.050	325
40 A or B	703	200	0.045	440
41 A or B	704	220	0.041	440
42 A or B	705	240	0.038	440
43 A or B	706	270	0.033	580
44 A or B	707	300	0.030	580

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-PRF-27/37,	MIL-PRF-27/38,	MIL-PRF-27/39,	MIL-PRF-27/40,
MIL-PRF-27/41,	MIL-PRF-27/42	MIL-PRF-27/44,	MIL-PRF-27/45,
MIL-PRF-27/46,	MIL-PRF-27/47,	MIL-STD-202	

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5950-2009-040)

Review Activities:
Army – CR4, MI
Navy – AS, MC, OS
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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